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Data Science

PLATEPAL

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Project Guide
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2. Introduction

Problem Identified:

- Food waste is a global crisis: 1/3 of all food produced globally is lost or wasted each year Retailers discard surplus food daily due to 'best before' dates, overstocking, and poor demand forecasting
- Food waste contributes ~8–10% of global greenhouse gas emissions Consumers lack easy access to affordable surplus food near them

Solution Proposed:

- PlatePal is a mobile-first food waste reduction platform that bridges the gap between food retailers with surplus inventory and cost-conscious, sustainability-aware consumers.
- Retailers list discounted 'surprise bags' of surplus food Consumers discover, reserve, and collect food near them Both parties track their real-world environmental impact Inspired by models like Too Good To Go, built for local communities

3. Objectives

1) To develop a mobile application that:Enables retailers to list surplus food discounted 'surprise bags' with pickup windows
2) Allows users to discover surplus food within a 5km radius via an interactive map
3) Provides a seamless in-app reservation system with pay-on-pickup functionality
4) Quantifies and displays the environmental impact (kg food saved, CO_2 prevented) for users and retailers
5) Reduces urban food waste and contributes to SDG 2, 12, 13, and 11

4)scope

●) Expanding the inventory listing system to support bulk uploads, automated expiry tracking, and demand forecasting tools for retailers.

●) Scaling beyond the 5km radius to cover wider urban and suburban areas, and eventually supporting multi-city or national rollout.

●) Moving from pay-on-pickup to full in-app digital payments, subscriptions, or wallet systems to reduce friction further.

●) Developing the Impact Dashboard into a full analytics suite — offering retailers detailed waste reduction reports, trends, and SDG compliance documentation.

●) Growing the Community Score leaderboard into a full social platform with badges, challenges, and neighbourhood-level competitions to drive engagement..

5. Feature /Functionality

- **Feature 1**:Retailers post 'surprise bags' of surplus food at discounted prices with a defined pickup window — quick, simple, and flexible.
- **Feature** 2An interactive map displays available food bags within a 5km radius, making discovery effortless for nearby users**.**
- **Feature 3:** Users reserve bags in-app and pay upon pickup, reducing digital payment friction while securing their order..
- **Feature 4:** Tracks total kg of food saved and CO_2 emissions prevented — giving real-time feedback to both users and retailers.

6. Outcome of Project

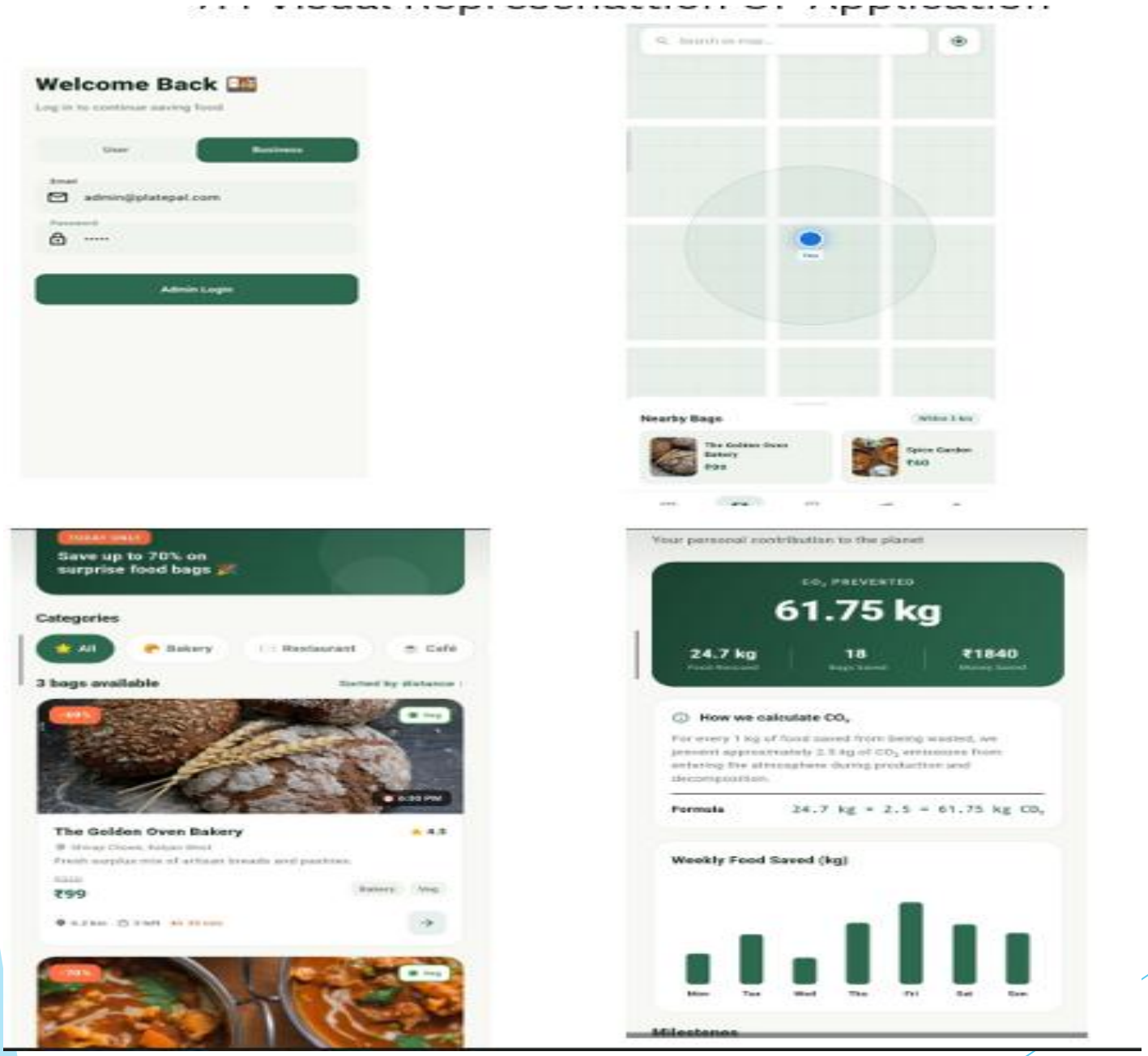
●)Reduced Food Waste — Diverts surplus retail food away from landfills through discounted surprise bags.
●)Lower Carbon Emissions — Reduces CO_2 and methane emissions tracked via the Impact Dashboard.
●)Better Food Access — Provides affordable food to budgetconscious consumers in local communities.
●)Retailer Revenue Recovery — Helps retailers recover partial income from unsold surplus inventory.
●)Community Engagement — Gamified leaderboard encourages habitual sustainable behaviour among users.
●)SDG Achievement — Directly contributes to SDG 2, 11, 12, and 13 through technology-driven food rescue.

7. Technology Stack

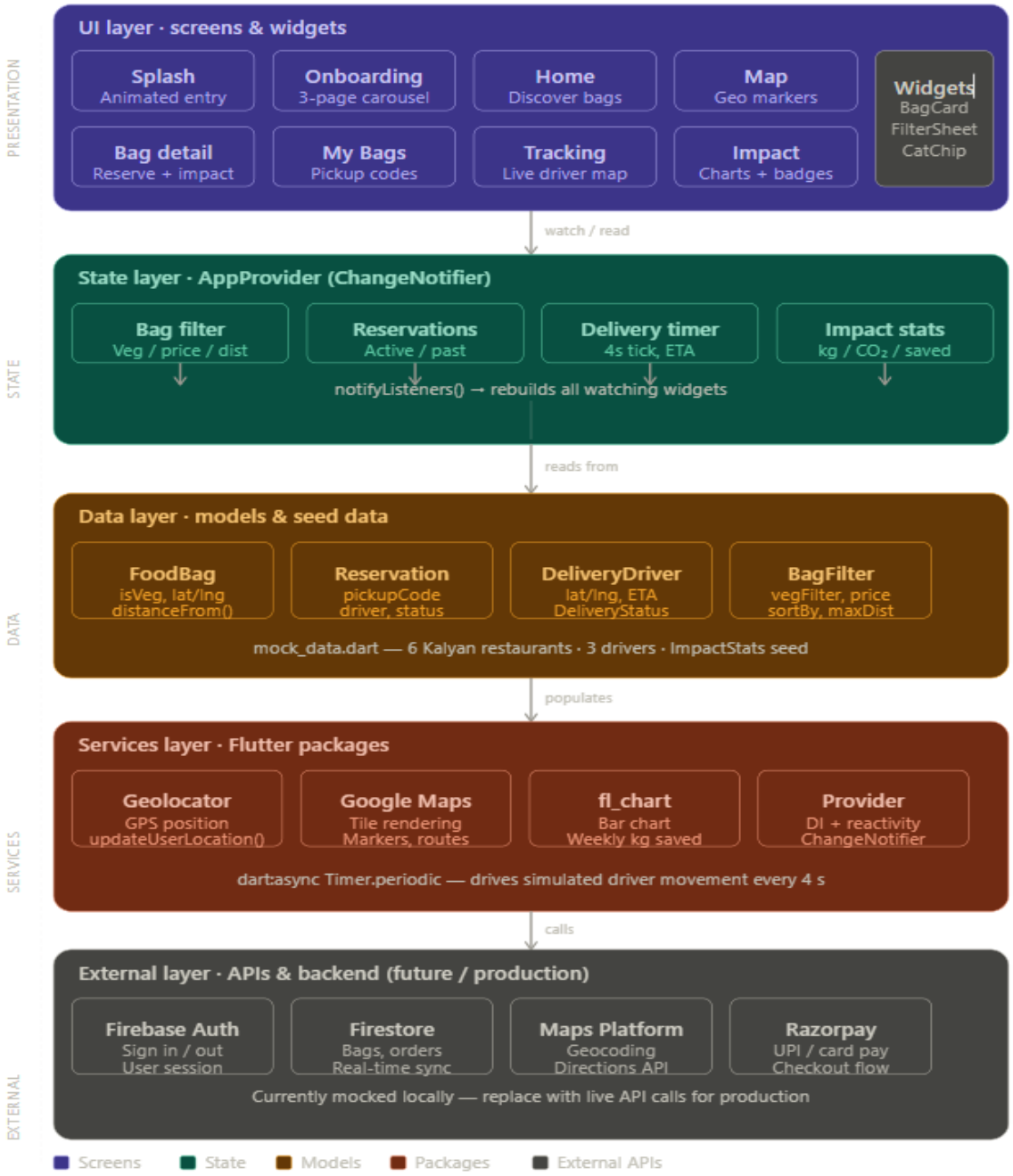
Frontend:
Flutter framework for building mobile UI (Android/iOS) Material Design for modern and user-friendly interface

Backend:
- Firebase Authentication for user login & security
- Data Management: It uses Local Persistence. The AppProvider.dart file uses shared_preferences to save and load data (like persisted_bags and persisted_reservations) as JSON directly on the device.
- Data Source: It relies on Mock Data (mock_data.dart) to simulate a database of retailers and food bags.

UI Mockups


Welcome Back
admin@platepal.com
Admin Login
Nearby Bags
Save up to 70% on surprise food bags
Categories
All
Bakery
Restaurant
Café
3 bags available
The Golden Oven Bakery
₹99
Your personal contribution to the planet
61.75 kg
24.7 kg
18
₹1840
How we calculate CO₂
Formula
24.7 kg × 2.5 = 61.75 kg CO₂
Weekly Food Saved (kg)
Milestones

8. Block Diagram



9)Conclusion

- Too Good To Go (Global) – Market leader; PlatePal differentiates via open-source architecture and community dashboard OLIO App (UK)

- Peer-to-peer sharing; PlatePal targets B2C retail surplus specifically Winnow (Restaurant AI) – Focuses on kitchen waste measurement; PlatePal focuses on distribution

Thank You...!!